# MI Net Growth Levels Management - Change Summary

Developer: Full Stack Developer | Branch: feature/mi-net-growth-levels | Date: November 2026

Implemented a full-stack system to manage MI Net Growth Levels for the SIA (Sales Incentive Automation) system using PostgreSQL, Django REST Framework, and React.js with full CRUD functionality, sequential NET_ID auto-generation, range contiguity/overlap prevention, file attachment uploads, and Role-Based Access Control (CFO, DGM, RM).

| No | File | Path | Used In | What Changed |
|---|---|---|---|---|
| 1 | **models.py** | `backend/growth_levels/models.py` | Database Table & ORM | Created SIA_MI_NET_GROWTH_LEVELS table with sequential NET_ID auto-generator, range validation, and overlap checks. |
| 2 | **views.py** | `backend/growth_levels/views.py` | REST API & Business Logic | Built MiNetGrowthLevelViewSet supporting CRUD, statistics, file download, and RBAC enforcement (CFO/DGM/RM). |
| 3 | **urls.py** | `backend/growth_levels/urls.py` | Backend Routing | Mapped /api/mi-net-growth-levels/ routes using DRF DefaultRouter for all endpoints. |
| 4 | **serializers.py** | `backend/growth_levels/serializers.py` | API Serialization | Created MiNetGrowthLevelSerializer with decimal validation, overlap checks, attachment URL builders, and audit user headers. |
| 5 | **pagination.py** | `backend/growth_levels/pagination.py` | Data Pagination | StandardResultsSetPagination supporting customizable page size and unpaginated query parameters. |
| 6 | **settings.py** | `backend/growth_backend/settings.py` | Backend Settings | Configured PostgreSQL with SQLite fallback, CORS headers, Media root and URL for file storage. |
| 7 | **tests.py** | `backend/growth_levels/tests.py` | Automated Testing | Added 11 unit tests covering CRUD, range overlap rejection (400), percentage limits (400), and RBAC permissions (403). |
| 8 | **seed_data.py** | `backend/seed_data.py` | Data Seeding | Populated initial enterprise growth tiers (NET001 to NET006) with realistic ranges, percentages, and sample attachments. |
| 9 | **docker-compose.yml** | `backend/docker-compose.yml` | PostgreSQL Setup | Containerized PostgreSQL 15 alpine database service listening on port 5432. |
| 10 | **api.js** | `frontend/src/services/api.js` | API Client Service | Axios client with multipart/form-data support, dynamic role headers, error message parser, and blob download. |
| 11 | **Navbar.jsx** | `frontend/src/components/Navbar.jsx` | Top Navigation | Enterprise navigation with title, simulator modal trigger, and live RBAC role switcher (CFO, DGM, RM). |
| 12 | **StatsCards.jsx** | `frontend/src/components/StatsCards.jsx` | Dashboard KPIs | Metric cards displaying Total Levels, Active Levels, Inactive Levels, Average Incentive %, and Max Range Ceiling. |
| 13 | **RangeVisualizer.jsx** | `frontend/src/components/RangeVisualizer.jsx` | Spectrum Analyzer | Visual range coverage bar confirming continuous non-overlapping active tiers from $0 to maximum ceiling. |
| 14 | **FilterBar.jsx** | `frontend/src/components/FilterBar.jsx` | Toolbar & Actions | Search by level or NET_ID, status filter tabs (All, Active, Inactive), CSV export button, and Add Level button. |
| 15 | **GrowthLevelTable.jsx** | `frontend/src/components/GrowthLevelTable.jsx` | Data Table | Table showing NET_ID, ranges, incentive %, active status pill, file download links, and role-based action buttons. |

## models.py

| No | Change | Details |
|---|---|---|
| 1 | **Table Name** | Created SIA_MI_NET_GROWTH_LEVELS table mapped through Django ORM db_table setting. |
| 2 | **NET_ID Field** | CharField unique=True (max_length=20) with automated sequence generator (NET001, NET002...). |
| 3 | **Range Fields** | LOWER_RANGE and UPPER_RANGE DecimalField(14,2) with non-negative and upper > lower constraints. |
| 4 | **Overlap Validation** | Model clean() verifies that no two Active records overlap in the lower_range to upper_range interval. |

## views.py (growth_levels)

| No | Change | Details |
|---|---|---|
| 1 | **CRUD Endpoints** | MiNetGrowthLevelViewSet provides POST (create), GET (list), GET // (retrieve), PUT/PATCH (update), DELETE (destroy). |
| 2 | **Search & Filter** | Implemented ?search= (Level, NET_ID, Comments) and ?status=Active\|Inactive query filtering. |
| 3 | **Statistics API** | Added /api/mi-net-growth-levels/statistics/ action calculating total, active, inactive, avg %, and max ceiling. |
| 4 | **RBAC Permissions** | check_role_permissions() blocks RM from all writes (403) and blocks DGM from delete actions (403). |

### views.py (growth_levels) - Continued

| No | Change | Details |
|---|---|---|
| 5 | **Download Attachment** | /api/mi-net-growth-levels//download_attachment/ streams binary file with Content-Disposition header. |
| 6 | **Multipart Support** | parser_classes = [MultiPartParser, FormParser, JSONParser] allows file uploads and JSON interchangeably. |
| 7 | **Ordering Logic** | Supported ?ordering= query parameter to dynamically sort by lower_range, percentage, or created_date. |
| 8 | **Unpaginated Export** | Supported ?paginate=false to return all records for reporting, visualizations, and client calculations. |

### urls.py

| No | Change | Details |
|---|---|---|
| 1 | **DefaultRouter Route** | router.register(r'mi-net-growth-levels', MiNetGrowthLevelViewSet) creates full REST URL tree. |
| 2 | **API Integration** | Mounted in growth_backend/urls.py under path('api/', include('growth_levels.urls')). |
| 3 | **Media File Handler** | Added static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT) to serve uploads in development. |

### serializers.py

| No | Change | Details |
|---|---|---|
| 1 | **Key Normalization** | to_internal_value() maps incoming uppercase keys (LEVEL, LOWER_RANGE) to lowercase model fields. |
| 2 | **Range Overlap Check** | Validates range bounds against database active records and returns clear 400 Bad Request if conflicts exist. |
| 3 | **Percentage Boundaries** | Enforces 0.00% <= percentage <= 100.00% returning field error if boundary is violated. |
| 4 | **Audit Header Tagging** | create() and update() auto-populate created_user and updated_user from request X-User-Role / X-User-Name. |
| 5 | **Attachment Metadata** | SerializerMethodFields dynamically expose attachment_name and absolute attachment_url. |

### pagination.py

| No | Change | Details |
|---|---|---|
| 1 | **Standard Pagination** | StandardResultsSetPagination with page_size=10, page_size_query_param='page_size', max_page_size=100. |
| 2 | **Metadata Response** | Returns count, total_pages, current_page, next, and previous links alongside data results. |

### docker-compose.yml

| No | Change | Details |
|---|---|---|
| 1 | **PostgreSQL 15 Container** | Configured postgres:15-alpine image with POSTGRES_DB=mi_net_growth_db on port 5432. |
| 2 | **Volume Persistence** | Mapped postgres_data volume to /var/lib/postgresql/data for local data durability. |

### api.js (Frontend Service)

| No | Change | Details |
|---|---|---|
| 1 | **Axios Client** | Created centralized Axios instance with baseURL (http://localhost:8000/api) and timeout. |
| 2 | **RBAC Interceptors** | Attaches current active role (X-User-Role) and user name from localStorage to every outbound request. |
| 3 | **Multipart Form Support** | growthLevelsApi.create() and update() automatically inspect data and apply multipart/form-data. |
| 4 | **Blob Download Helper** | downloadAttachment() requests responseType: 'blob' and generates temporary DOM anchor to download. |
| 5 | **DRF Error Formatter** | formatApiError() unpacks DRF error dictionaries into human-readable toast notification strings. |

### Frontend Components (React.js)

| No | Change | Details |
|---|---|---|
| 1 | **Navbar.jsx** | Displays system title, simulator modal shortcut, and live role switcher (CFO, DGM, RM). |
| 2 | **StatsCards.jsx** | 5 clean metrics cards showing Total Levels, Active Levels, Inactive Levels, Avg Incentive %, Max Range. |
| 3 | **RangeVisualizer.jsx** | Visual tier coverage bar displaying continuous non-overlapping distribution from $0 to ceiling. |
| 4 | **FilterBar.jsx** | Search by level or NET_ID, status filter tabs (All, Active, Inactive), CSV export, and Add Level button. |
| 5 | **GrowthLevelTable.jsx** | Table displaying NET_ID, Level, Ranges, Incentive %, Active status pill, Attachment download, and actions. |
| 6 | **LevelFormModal.jsx** | Modal with client-side range checks, real-time overlap collision alert, and file upload input. |
| 7 | **DeleteConfirmModal.jsx** | Confirmation dialog showing tier details before deletion, restricted to CFO. |
| 8 | **TierDetailDrawer.jsx** | Slide-over drawer displaying full tier specs, payout scenarios, audit trail, and document downloader. |
| 9 | **Toast.jsx** | Toast notification container providing real-time feedback for create, update, delete, and validation errors. |

### SimulatorModal.jsx

| No | Change | Details |
|---|---|---|
| 1 | **Deal Volume Input** | Allows entering gross net growth amount with quick presets ($25k, $100k, $250k, $500k, $1M). |
| 2 | **Bracket Matching** | Evaluates active tiers to find matching qualification tier and applicable incentive percentage. |
| 3 | **Formula Breakdown** | Displays exact formula calculation and distance needed to unlock the next higher tier bracket. |

### .env (Backend & Frontend)

| No | Change | Details |
|---|---|---|
| 1 | **USE_POSTGRES** | USE_POSTGRES=False (defaults to SQLite for instant local dev) or True to connect to PostgreSQL. |
| 2 | **DB Connection Keys** | DB_NAME=mi_net_growth_db, DB_USER=postgres, DB_PASSWORD=postgres, DB_HOST=localhost, DB_PORT=5432. |
| 3 | **VITE_API_URL** | VITE_API_URL=http://localhost:8000/api for Vite React client. |

### User Flow

| No | Flow | Steps |
|---|---|---|
| 1 | **View & Search Tiers** | User navigates to dashboard, views KPI metrics and active range spectrum, searches by level name or NET_ID, and filters by status (All, Active, Inactive). |
| 2 | **Add New Tier (CFO/DGM)** | User clicks 'Add Level', fills level title, lower & upper ranges, percentage, comments, and optional file. Client and backend validate range contiguity and overlap before committing. |
| 3 | **Edit Tier (CFO/DGM)** | User clicks Edit icon on a row, form pre-populates with existing data. User modifies range or replaces attachment. Table and KPI stats auto-refresh upon save. |
| 4 | **Delete Tier (CFO Only)** | CFO clicks Delete icon, confirmation dialog displays record summary. Deletion permanently removes tier. DGM and RM users have Delete button disabled (HTTP 403 on API). |
| 5 | **Simulate Deal Payout** | User clicks 'Simulator' in header, inputs gross deal volume (e.g. $250,000), engine highlights matching tier, applicable percentage (8.75%), and calculates payout ($21,875.00). |

### Testing Checklist

| No | Test Case | Expected Result |
|---|---|---|
| 1 | **Database Table Creation** | Table SIA_MI_NET_GROWTH_LEVELS created in PostgreSQL via makemigrations and migrate. |
| 2 | **Auto NET_ID Generation** | Sequential ID (NET001, NET002, ...) auto-generated on record creation without manual input. |
| 3 | **List Records API (GET)** | Returns HTTP 200 OK with paginated list, count, total_pages, and results array. |

## Testing Checklist - Continued

| No | Test Case | Expected Result |
|---|---|---|
| 4 | **Single Record Retrieval (GET)** | Returns HTTP 200 OK with single record details matching requested PK ID. |
| 5 | **Multipart File Upload (POST)** | Returns HTTP 201 Created; uploaded file stored in media directory and attachment URL returned. |
| 6 | **Range Overlap Validation** | Attempting to create or update an overlapping active range returns HTTP 400 Bad Request with conflicting tier names. |
| 7 | **Percentage Limit Validation** | Percentage < 0 or > 100 returns HTTP 400 Bad Request. |
| 8 | **Upper > Lower Range Validation** | Upper range <= lower range or negative lower range returns HTTP 400 Bad Request. |
| 9 | **RM Read-Only Access (RBAC)** | RM role sending POST, PUT, PATCH, or DELETE receives HTTP 403 Forbidden. |
| 10 | **DGM Delete Prevention (RBAC)** | DGM role sending DELETE receives HTTP 403 Forbidden. Can execute POST, PUT, and PATCH. |
| 11 | **CFO Full Permissions (RBAC)** | CFO role can create, view, update, and delete growth levels. |
| 12 | **Attachment File Download** | GET download_attachment streams file with Content-Disposition header and correct filename. |
| 13 | **Frontend Table & Auto-Refresh** | Table renders active pills, formatted currency, and auto-refreshes immediately after mutations. |
| 14 | **Automated Backend Test Suite** | All 11 unit tests in tests.py pass successfully (Ran 11 tests in 0.101s, OK). |

## Git Status & Commit Log

**Branch:** feature/mi-net-growth-levels | **Commit:** 4e7d294 | **Author:** Full Stack Developer
**Commit Message:** feat(mi-net-growth): complete full-stack implementation with Django ORM, REST API, React frontend, and Postman suite

**Tracked Files in Repository (Clean Working Tree):**
- backend/growth_levels/models.py ..... [SIA_MI_NET_GROWTH_LEVELS table model]
- backend/growth_levels/views.py ..... [MiNetGrowthLevelViewSet CRUD & RBAC]
- backend/growth_levels/serializers.py ..... [DRF serializer with overlap validation]
- backend/growth_levels/urls.py ..... [API routes registration]
- backend/growth_levels/pagination.py ..... [Standard pagination handler]
- backend/growth_levels/tests.py ..... [11 unit tests passing]
- backend/growth_backend/settings.py ..... [PostgreSQL / SQLite config & CORS]
- backend/seed_data.py ..... [Database seeder for sample tiers]
- backend/docker-compose.yml ..... [PostgreSQL 15 service]
- frontend/src/services/api.js ..... [Axios CRUD service & multipart handler]
- frontend/src/components/Navbar.jsx ..... [Header & RBAC role switcher]
- frontend/src/components/StatsCards.jsx ..... [Executive KPI cards]
- frontend/src/components/RangeVisualizer.jsx ..... [Contiguity spectrum analyzer]
- frontend/src/components/FilterBar.jsx ..... [Search & status filter toolbar]
- frontend/src/components/GrowthLevelTable.jsx ..... [Data table with action triggers]
- frontend/src/components/LevelFormModal.jsx ..... [Add/Edit modal with overlap check]
- frontend/src/components/DeleteConfirmModal.jsx ..... [Deletion confirmation dialog]
- frontend/src/components/SimulatorModal.jsx ..... [Net revenue payout simulator]
- frontend/src/components/TierDetailDrawer.jsx ..... [Tier specs & dossier slide-over]
- frontend/src/components/Toast.jsx ..... [Toast notification system]
- postman/MI_Net_Growth_Levels.postman_collection.json [Postman collection v2.1.0]
- postman/POSTMAN_GUIDE.md ..... [Complete API testing guide]
- run_all.ps1 ..... [One-click startup PowerShell script]